

Differentiation Before Frequency: Survival, Variation, and Internal Evaluation in Language

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Abstract

Languages routinely generate structurally valid variants, yet derivation alone does not guarantee their survival in use. This essay develops an account of how linguistic forms survive, spread, or are reshaped in use by examining a set of concrete cases, primarily from Telugu plural morphology but extending across registers, dialect contact, and cross-linguistic borrowing. Through a series of illustrations, I show that grammatically possible alternatives are not neutral once enacted in use. Instead, they diverge systematically according to how readily they integrate into ongoing articulatory and structural coordination. Building on this observation, I propose internal evaluation as a context-sensitive process of differentiation operating during speech enactment. By identifying this locus of differentiation, this study reframes variation, regional restriction, and linguistic change as outcomes of coordination-based differentiation during use, rather than as effects of frequency or conscious selection alone. In doing so, it clarifies a persistent explanatory gap in linguistic theory.

Keywords: Internal evaluation, Differential stabilization, Frequency effects, Production enactment, Structural organization, Language variation

1. Introduction: When “Correct” Forms Do Not Survive

Languages are often taught as systems of rules. If a form follows the rules, it is considered correct, and if it violates them, it is incorrect. This assumption is so deeply ingrained that we rarely stop to question it. Yet, everyday speech constantly presents a quiet contradiction, which is that not all correct forms survive, and not all surviving forms are equally accepted across contexts. Anyone who listens to a living language notices this pattern. Some words feel natural and effortless, while others, though perfectly derivable, sound awkward, heavy, or strangely artificial. Speakers may understand them, but hesitate to use them. Over time, such forms either retreat into formal registers, survive only in poetry, or disappear altogether. In other words, grammatical correctness alone does not predict whether a form will actually circulate in everyday use.

This raises a simple but unsettling question: what determines whether a form actually lives in a language? Let us consider plural formation in Telugu. Grammar provides a clear mechanism for this account where a noun combines with a plural marker, and the result should be acceptable. On paper, the process is straightforward and productive. Yet in practice, speakers strongly prefer some plural forms and avoid others, even when both are structurally possible. The crucial part is not that speakers make mistakes or simplify forms. Rather, they consistently converge on certain outcomes while quietly rejecting others. The rejected forms are not unintelligible, nor are they ungrammatical in a strict sense. They simply fail to circulate in regular use.

This difference, between what can be formed and what can be stabilised, is rarely addressed directly. Traditional grammatical theory has long distinguished between competence and performance by effectively placing questions of usage outside the explanatory scope of grammar itself (Chomsky, 1965). Later work has acknowledged that grammatical systems alone cannot account for patterns of frequency, entrenchment, or long-term survival (Newmeyer, 1983). Usage-based and emergent approaches have gone further, arguing that grammatical structure arises from repeated use rather than preceding it (Bybee, 2010; Hopper, 1987). These approaches successfully explain how repetition, frequency, and distribution shape linguistic structure. Yet when similar patterns recur across morphology, dialects, and even across unrelated languages, it becomes harder to dismiss them as incidental.

A useful way to think about this is to separate derivation from survival. Derivation answers the question: *Can this form be produced by the rules?* But, survival answers entirely a different question: *Will speakers actually keep using it?* This separation is important because languages are full of forms that pass only the first test but fail the second. They may appear in textbooks, dictionaries, or highly formal registers, yet remain absent from natural speech. Their absence is not random, but reflects a collective judgment that emerges without instruction or explicit agreement. What is striking is how consistent this judgment is. Entire speech communities independently converge on these same outcomes without explicit coordination. This consistency suggests that stabilisation of forms cannot be explained by rule-based derivation alone.

One reason grammar struggles to explain this phenomenon is that it treats language as primarily symbolic. Words are modelled as combinations of abstract units governed by formal rules. Yet language is not only symbolic, it is also physical. Every utterance is a physical event. It is produced, transmitted, and perceived through biological systems. Research in phonetics and speech perception emphasised that linguistic forms are experienced as acoustic and articulatory events, not merely computed as symbolic structures (Browman & Goldstein, 1992; Fowler, 1986). The auditory system is sensitive to rhythm, timing, weight, pressure, and balance. As a result, some sequences integrate smoothly into ongoing speech, while others introduce greater coordination demands, even though both satisfy grammatical constraints. Speakers do not typically calculate syllable counts or articulatory effort when choosing a form. Instead, such pressures operate continuously during speech production and perception (Lindblom 1990b; Ohala 1993). Over time, such pressures are associated with patterns of reduction and differential recurrence (Lindblom 1990a; Kirchner 1998), which in turn

contribute to frequency divergence and representational entrenchment (Bybee 2001; 2010; Pierrehumbert 2001; Wedel 2006). These perspectives have greatly advanced our understanding of how linguistic patterns emerge and stabilise over time. However, they primarily describe how usage patterns become amplified once certain forms begin to recur.

Grammar can describe what is possible, and pressure can describe tendencies in realisation and frequency amplification. Yet neither directly explains why some grammatically possible alternatives begin to circulate more readily than others in the first place. This shared limitation reveals an explanatory gap. If grammatical correctness does not guarantee survival and pressure alone does not determine stabilisation, what mechanism differentiates among grammatical rivals during use? Similarly, usage-based and exemplar accounts further explain how repetition and frequency consolidate emerging patterns, but they leave open the related question of how differentiation first emerges among co-possible forms at the point of use, before frequency divergence and social interpretation take hold. This question becomes especially visible when similar patterns operate across everyday speech and poetry, across dialect boundaries, and even across unrelated languages with no shared history. When such patterns of structural tension recur, languages respond in familiar ways through compression, expansion, or other forms of structural adjustment. These responses are patterned rather than random innovations. Therefore, any adequate account of linguistic stabilisation must address not only how usage patterns become amplified over time, but also how differentiation first emerges among co-possible grammatical alternatives during use.

This essay does not begin with a theory. It begins with a set of recurring tensions that grammar leaves unresolved, such as why some words feel complete while others feel unfinished; why the same grammatical element behaves differently depending on the word it attaches to, the register it appears in, or the neighbouring language system it encounters. The sections that follow trace the same patterns across morphology, registers, dialect contact, lexical competition, distant languages, and moment-to-moment decisions in everyday speech. Through a small set of concrete cases such as plural formation, poetic usage, dialectal variation, and cross-linguistic borrowing, the aim is to first establish how persistently they surface across domains. The purpose is to ask whether these recurring patterns point to a more specific locus of differentiation during language use itself. Once their consistency is demonstrated, the later sections return to these observations in order to consider what they suggest about processes that differentiate among co-possible forms in language use.

2. The Same Suffix, Different Fates

One of the most revealing places to look for hidden structure in a language is not in rare or exotic constructions, but in the most ordinary ones. Elements that appear simple and well-understood often expose the deepest constraints when examined closely. In Telugu, plural formation offers exactly such a window. At first glance, the plural marker -lu (/lu/) appears straightforward. It is taught as a general plural suffix, attachable to nouns to indicate multiplicity. From a grammatical point of view, its behaviour seems predictable and regular.

Yet when we observe how -lu actually behaves in living usage, a more complex picture emerges. When the same suffix attached to different singular forms, it often produces sharply different outcomes. Some plural forms stabilise effortlessly and feel native. Others remain marginal, restricted to specific registers. Still others, though morphologically well-formed in principle, fail to gain traction altogether. The suffix itself remains constant. What changes, however, is how the resulting form integrates into the language system.

Let us consider a basic singular noun like *ceṭṭu* /tʃeṭṭu/ ('tree'). When the plural marker -lu is applied, it does not yield a single stable form, but several competing ones such as *ceṭlu* /tʃeṭlu/, *ceṭṭulu* /tʃeṭṭulu/, and *ceṭṭul* /tʃeṭṭul/, all of which are derivationally plausible within the morphophonological resources of the language. Yet speakers do not treat these outcomes as equivalent. One form (*ceṭlu*) has emerged as the most widely used plural, functioning comfortably in everyday and academic contexts. Another retains a more expanded shape and remains available across some regional varieties of speech. A third survives selectively, but only under special conditions. Each outcome involves the same plural marker -lu, but the resulting words shapes differ. Observations from conversational usage and consultant reports consistently show that one form circulates freely across informal speech, while others remain regionally or stylistically restricted.

This pattern suggests that plural formation may not be merely a matter of suffixation. It involves integration. The suffix must align with the existing word shape in ways that speakers find acceptable. Where such alignment is strong, a form stabilises. Where it is weaker, forms remain limited in their distribution, even when they are grammatically legitimate. Morphological theory has acknowledged that affixes do not behave uniformly across lexical items, and that paradigms exert pressure on the realisation of forms (Aronoff 1976; Blevins 2006). However, these accounts typically describe the outcomes rather than explain why certain integrations succeed while others do not.

Grammar often presents suffixes as uniformly productive. If a suffix can attach to a category of words, it is assumed to do so freely. Natural languages rarely behave so cleanly. What Telugu shows us is that productivity is filtered. The suffix -lu does not act identically across all nouns. Its success depends on how the resulting form aligns with broader preferences of the language such as timing, structural balance, and overall word-shape. Research on usage-based morphology has repeatedly shown that frequency, repetition, and ease of articulation influence which forms become entrenched (Bybee, J. L. 1985; Hay 2001). However, these correlations do not by themselves explain why one form becomes dominant among co-possible alternatives at the point of use. Over time, these intuitions shape usage, and usage reshapes the language. The result is that plural formation appears rule-governed on the surface, but selective in practice.

A useful way to understand these differences is to think in terms of compression and expansion. Some plural forms compress neatly into a tight rhythmic unit. They maintain a balance in consonant-vowel structure and resolve cleanly. These forms tend to circulate widely. Other forms expand in ways that disrupt this balance. In such cases, the meaning may be clear, but the form feels less economical. Such expansions are not necessarily rejected outright, but they often struggle to become the default choices. Historical morphology and phonology

document similar tendencies toward restructuring in high-frequency domains (Hock 1991; Bybee, J. 2015). This has nothing to do with conscious preference or stylistic taste. Speakers are not selecting shorter forms because they are lazy, nor avoiding longer ones because they are incorrect. The distinction concerns how forms integrate into the larger dynamics of speech.

But then what is driving this selective behaviour, whereby some forms stabilise while others remain marginal despite being morphologically licensed? This selective behaviour of the plural suffix becomes clearer when we examine cases where grammatical explanations reach their limits.

2.1 *paṇḍu* → *paṇḍlu* / *paḷḷu*, but not *paṇḍulu*

From a grammatical point of view, nothing in the plural template appears to prevent the formation *paṇḍulu* /paṇḍulu/. The singular *paṇḍu* /paṇḍu/ ('fruit') ends in a vowel, and the plural suffix -lu should attach without difficulty. Yet in natural usage, *paṇḍulu* remains marginal in conversation usage and is typically avoided in rapid speech. Instead, *paṇḍlu* predominates, and in many contexts compresses further to *paḷḷu* /paḷḷu/. What is striking here is not the existence of alternation, but its direction. The language consistently moves toward compression rather than expansion. The cluster formation in *paṇḍlu* and further assimilation in *paḷḷu* reduce syllabic weight and produce a tighter consonantal alignment than the expanded *paṇḍulu*. This mirrors broader historical tendencies, where forms repeatedly reduce rather than accumulate material over time. Comparable compression patterns are observable in English high-frequency speech, where "want to" routinely contracts to 'wanna' under conditions of rapid articulation, illustrating how repeated articulatory alignment favours structurally tighter word structures. The rejected form (*paṇḍulu*) is not unclear or incorrect. It simply fails to circulate widely. This suggests that the plural formation is sensitive to structural form, not merely to morphological transparency.

2.2 *pannu* splits: *paḷḷu* and *pannulu*

An even more revealing case of the selective behaviour involves the surface form *pannu* /pannu/. The word can mean 'tooth' or 'tax', depending on context, yet their plural outcomes diverge sharply. For 'tooth', speakers say *paḷḷu* /paḷḷu/, not *pannulu* /pannulu/. For 'tax', the preferred plural is *pannulu*, not *paḷḷu*. This contrast cannot be explained by phonology alone, since the singular form is identical. Nor can it be explained by semantics in any simple sense. What differs is the usage domain in which each meaning circulates. Corpus observations suggest that the 'tooth' sense overwhelmingly occurs in conversational contexts, while the 'tax' sense is concentrated in formal and institutional discourse.

The "tooth" sense belongs to a high-frequency, embodied domain. Its plural compresses aggressively, aligning with a lighter plural pattern. The "tax" sense, by contrast, is abstract, institutional. Its plural tolerates expansion and remains closer to the morphological template. Studies of lexical diffusion and paradigm uniformity show that frequency and domain of use can condition morphological outcomes (Phillips 1984; Bybee, J. 2007). What we observe here

is interaction between word structure and domain of circulation. In this instance, the language does not apply the plural marker mechanically. Its outcome depends on how the resulting form fits the context in which it recurs.

2.3 Why *ēḍu* avoids *ēḍulu*

A final puzzle sharpens the point further. The word *ēḍu* /e:ḍu/ ('year') does not normally pluralise as *eḍulu* /e:ḍulu/, or *ēḍlu* /e:ḍlu/, by analogy with forms such as *paṇḍlu*, even though both plural forms would be grammatically straightforward. Instead, speakers use forms like *ēṇḍlu* /e:nḍlu/ or *ēḷḷu* /e:ḷḷu/. Again, the rejected forms are fully intelligible. Yet both forms sound constructed rather than established. In contrast, the accepted forms involve internal restructuring rather than simple suffixation. The insertion of a nasal (*ēṇḍlu*) or lateral assimilation (*ēḷḷu*) produces a more compact consonantal sequence and avoids the expanded vowel sequence that *eḍulu* would create. Here the pattern shows reorganisation over transparent addition.

Across these observations, a pattern emerges. Forms that expand the word by adding syllabic weight tend to struggle to stabilise. Forms that compress or reorganise the word tend to circulate more widely. This tendency appears repeatedly in the behaviour of Telugu plurals, from *paṇḍu* → *paṇḍulu* → *paḷḷu*, to the divergence between *paḷḷu* ('teeth') and *paṇṇulu* ('taxes'), and the restructuring of *ēḍu* into *ēṇḍlu* or *ēḷḷu*. The same suffix behaves differently depending on the word, its usage domain, and its phonological environment. Identical singular forms can pluralise differently when their usage histories diverge. None of this follows automatically from grammatical rules, yet none of it is random. Speakers converge consistently on the similar outcomes across contexts. This plural formation therefore reveals that grammatical derivation does not guarantee distributional success.

It is tempting to treat such plural forms as exceptions tied to particular nouns. But such an argument misses the broader point. The suffix -lu works productively in countless cases. What varies is the environment into which it is introduced. When the suffix integrates smoothly into the word's structure, the result circulates widely. When it disrupts the shape, the result remains restricted or reshaped. The language does not treat the suffix in isolation. It differentiates among the full forms that emerge after suffixation. Once this pattern is recognized, similar asymmetries become visible beyond plural formation, across other morphological processes. Certain combinations feel inevitable while others remain peripheral.

At this stage, it is sufficient to establish that such patterned differentiation exists. The behaviour of the -lu suffix shows that grammatical possibility does not by itself determine distribution. Plural formation, often treated as routine, turns out to be a sensitive diagnostic site for observing how languages regulate form beyond rule alone. If grammatical possibility is not sufficient for survival, then what differentiates among co-possible derivations?

3. Selective Survival Across Registers Within a Single Language

Up to this point, we have seen that grammatical derivation alone does not determine which forms survive in a language. The behaviour of the plural suffix -lu already shows that forms permitted by the grammar do not circulate equally, with some stabilising while others quietly fall away. This pattern becomes even clearer when we look not across different languages, but within the same language, across different registers such as everyday speech, poetry, and formal usage. What emerges is not merely stylistic variation, but systematic differences in how the same forms function under different conditions.

In everyday spoken Telugu, language operates under constant demands of speed, interaction, and real-time coordination. Words must fit smoothly into ongoing speech without disrupting timing or articulatory coordination. In this environment, forms are shaped by economy and flow. Plural forms such as *ceṭlu*, *paṇḍlu*, or *paḷlu* that compress neatly, resolve quickly, and move efficiently through articulation tend to recur effortlessly. Forms that introduce extra syllabic weight such as *ēḍulu* or *ceṭṭulu*, often remain limited in spontaneous speech, not because they are unintelligible or incorrect, but because they introduce additional structural load. This pattern aligns with observations from speech production studies showing that sequences which integrate smoothly into ongoing articulation are more likely to recur, while those that interrupt planning or execution tend to be avoided (Levelt 1989; Lindblom 1990b).

Poetry is often described as a domain of freedom or license, but this description is misleading. Poetic language does not relax constraints, it reorganises them. Where everyday speech favours compactness and efficiency, poetry operates under metrical and rhythmic demands. This helps explain why certain plural forms rejected in conversational usage may appear more naturally in verse. A form such as *ceṭṭul*, for example, compresses into a two-beat structure while retaining a strong consonantal core. The geminate consonant provides acoustic weight, allowing the word to anchor a metrical position within the line. The same form that may introduce excess weight in everyday speech can align more effectively within poetic rhythm. Its distribution reflects structural suitability within a different set of constraints (Fabb 1997).

Formal or bookish usage introduces yet another set of conditions. Here, forms may persist not because they integrate naturally into speech or rhythm, but because they are preserved through education, standardisation, and textual transmission. In formal written registers, plural forms such as *pāṭhaśālalu* /pa:ṭʰaʃa:lalu/ ('schools') or *vāhanālu* /va:ʱana:lu/ ('vehicles') are treated as standard, while conversational speech frequently favours borrowed or colloquial alternatives such as *schoolu* /sku:ḷlu/ or *baṇḍlu* /baṇḍlu/. These formal plural forms persist as part of a learned system, yet circulate less frequently in everyday interaction. This distribution aligns with broader patterns of register stratification documented in sociolinguistic research on standardisation (Milroy & Milroy 1999).

Across these three registers, the grammatical system itself remains constant. What changes are the conditions under which forms are used. A form that circulates widely in conversation may be unfavoured in poetry, and a form preserved in formal writing may remain peripheral in

everyday speech. Therefore, registers are not merely stylistic layers, they function as distinct contexts in which forms encounter different structural demands. Speakers shift between these environments with ease, yet the patterns of distribution remain consistent within each domain. These contrasts suggest that stabilisation is sensitive to context-specific conditions rather than governed solely by grammatical possibility.

Although register-based variation is often treated as secondary to grammar, the evidence that surfaced through these registers points in a different direction. Registers do not merely decorate grammatical structure, but they shape which forms circulate broadly, which remain specialized, and which are confined to particular domains. Across these registers, the grammatical system itself remains constant, yet its outcomes differ depending on the conditions under which forms are used. If neither the grammar nor lexical items change, what then explains the fact that a form may settle in one register but not in another? Is this pattern restricted to morphology and registers, or does it also surface in dialectal contact?

4. Across Borders: Dialect, Contact, and Structural Repair

So far, we have seen that grammatical derivation does not guarantee distribution. Some forms circulate widely across registers, while others remain limited, even when the grammatical system itself remains unchanged. When we turn to dialect contact, a similar pattern emerges, but with an important difference. Instead of merely restricting forms to particular domains, contact settings often reshape them. This reshaping is not creative in the ordinary sense. It is structural.

In some Telugu-speaking regions, particularly those closer to Tamil-speaking areas, speakers commonly say *kōsaram* /ko:saram/ where Standard Telugu uses *kōsam* /ko:saam/, meaning ‘for’ or ‘for the sake of’. From a prescriptive point of view, *kōsaram* is often dismissed as incorrect or rustic. Yet its distribution is neither random nor careless. The form appears most commonly in regions where Telugu comes into sustained contact with a phonological system shaped by Tamil. The significance of this shift lies not in meaning, which remains unchanged, but in form.

The form *kōsam* ends in a fricative-plus-nasal sequence that is perfectly acceptable in Standard Telugu. However, in speech communities influenced by Tamil phonology, this ending that does not align as smoothly with preferred word final patterns. Tamil phonology strongly favours vowel-supported endings and restricts certain coda clusters, often producing disyllabic shapes such as *-aram*, or *-uram*. As a result, *kōsam* may present a less stable word-final contour in such environments. When such misalignment persists, the form does not simply remain variable or disappear. Instead, it is reshaped until it reaches a configuration that integrates more comfortably within the receiving phonological system. This is in line with findings in contact linguistics where boundary-crossing forms are frequently reorganised to conform to the phonological patterns of the recipient system (Thomason & Kaufman 1988).

Crucially, speakers are not typically modifying the word through explicit deliberation. No one announces the addition of an extra segment, yet the repair follows systematic paths. It

would be insufficient to treat such changes as casual drift, since their recurrence across speakers suggests patterned restructuring rather than sporadic alteration. If this were mere imitation or error, we would expect wide variation. Instead, we observe consistent outcomes across speakers. Research on contact-induced change has repeatedly shown that adaptation tends to follow structural properties of the recipient language, rather than the phonetic details of the source (Heine & Kuteva 2005; Matras 2009). A defining feature of such restructuring is that meaning remains stable. If we observe closely, the function of *kōsam* is preserved entirely in *kōsaram*. The adaptation affects only the structural form while preserving semantic function. This suggests that reshaping addresses structural fit rather than communicative clarity. Comparable observations appear in studies of phonological adaptation, where borrowed forms retain semantic identity while undergoing significant reorganisation (Kang 2011).

That this is not an isolated phenomenon becomes clearer when we observe that similar expansions have entered Standard Telugu. The form *avasaram* /avasaram/ (‘necessity’), ultimately traceable to Sanskrit *avaśyam* /avaɐjam/, which underwent structural reshaping during its incorporation into Telugu. But unlike *kōsaram*, which remains regionally distributed, *avasaram* eventually stabilised across speech communities and became part of Standard Telugu. Historical linguistics provides many such examples in which forms shaped at contact boundaries migrate inward once their restructured shape proves compatible across wider usage (Hock 1991).

With these illustrations, the pattern now becomes more visible. When distinct phonological systems come into contact, structural tension becomes visible, and when such tension persists, reshaping frequently follows. The regularity with which unsettled forms are reorganised rather than abandoned echoes patterns already observed in morphology and registers. It is this consistency of outcomes across such different linguistic environments that invites further explanation.

5. Lexical Competition Beyond Affixation

The patterned differentiation observed in morphological reshaping is not limited to affixation. Similar patterns also appear in cases of lexical rivalry, where semantically equivalent forms exhibit divergent distributional trajectories. Consider the lexical equivalents for ‘pillow’ such as *talagaḍa* /talagaḍa/, *metta* /met:a/, and *dinḍu* /dinḍu/. Although all three forms denote the same referent and are grammatically licensed within the language, their distribution is not uniform. While *dinḍu* circulates widely in everyday speech, the others tend to remain regionally or stylistically restricted. No semantic contrast or grammatical distinction accounts for this divergence.

Structurally, however, the forms differ in syllabic span and internal configuration. *talagaḍa* extends across four light syllables, *metta* contains a medial geminate interruption, whereas *dinḍu* forms a compact disyllable with internal nasal-stop closure. These contrasts suggest that lexical rivalry may not be resolved solely at the level of derivation or meaning, but may instead reflect how different word shapes function across recurring environments of use. On this view, broader distribution does not follow automatically from semantic transparency

or grammatical legitimacy. Seen together with the cases discussed across morphology, registers, dialect contact, this example raises a broader question: if multiple grammatical and semantically equivalent forms coexist, what determines which circulate widely in everyday use and which remain confined?

6. Distant languages, Similar Repairs

Up to this point, all examples have remained within a single language. We have examined morphological alteration, register-based differentiation, dialect contact, and lexical rivalry, but always within Telugu and its immediate linguistic environment. A skeptical reader might still argue that what we are seeing is a set of language-specific tendencies internal to Telugu and its immediate neighbours, rather than more general structural patterns. That explanation becomes increasingly difficult to maintain once we step outside this space altogether.

For instance, when English words enter Japanese, they undergo systematic reshaping. Consonant clusters are broken apart, vowels are inserted, and word-final consonants are avoided. Japanese syllable structure is predominantly CV-based and mora-timed, with strong restrictions on complex codas and consonant clusters. As a consequence, the English word *strike* becomes *sutoraiku* /sutoraiku/ in Japanese. Such reorganisation is often explained as a straightforward consequence of phonotactic limitation as Japanese does not permit certain clusters, so vowels are inserted to make pronunciation possible. While this description captures part of the process, it leaves an important question unaddressed. Phonotactic restrictions explain why *strike* cannot remain unchanged, but it does not by itself explain why a specific reshaped form ‘*sutoraiku*’ stabilises and circulates consistently across speakers rather than remaining variable. The borrowed form does not linger indefinitely in competing states. It settles into a pattern that aligns with Japanese syllable structure and timing.

This reshaping is neither arbitrary nor exploratory. Japanese speakers do not attempt to preserve the fine phonetic details of the English source, nor do they generate unlimited alternatives. Instead, the resulting form consistently conforms to Japanese phonological patterns. Consonants are separated by vowels, syllables align with preferred timing units, and the word is given a stable ending that integrates smoothly into Japanese speech. Studies of loanword phonology have shown that such adaptations follow the internal phonological patterns of the recipient language rather than the phonetic surface of the source (Kang 2011). This parallels what we observed earlier with *kōsam* becoming *kōsaram*. The source form is not rejected outright. It is reorganised in ways that conform to the receiving system.

What makes this comparison especially striking is that Japanese and Tamil, languages with no historical relationship, display remarkably similar outcomes when confronted with structurally unsettled forms. Both languages favour predominantly open syllable structures and restrict complex word-final clusters. Japanese enforces a moraic timing system with limited codas, while Tamil similarly constrains word-final clusters and often resolves them through vowel-supported endings. As a result, when confronted with foreign or unsettled forms, both languages converge on similar solutions such as vowel insertion and syllabic expansion. This

resemblance cannot be attributed to shared ancestry or borrowing between languages. It arises independently guided by the structural preferences of each system.

Once placed side by side, these challenge explanations based solely on habit or imitation. The outcomes are too consistent and too system-specific. Unrelated languages respond to structural misalignment in ways that are predictable within each system, yet strikingly similar in type across systems. Instead of rejecting unfamiliar forms, they reshape them in directions compatible with their phonological patterns. Comparative studies of loanword adaptation consistently show that the direction of change is guided by recipient-language structure rather than chance (Itô & Mester 1995).

These observations emphasise that reshaping is not confined to one language or one region. It appears as a recurring response to structural tension, whether triggered by register shift, dialect boundary, or foreign input. What varies from language to language is not whether reshaping occurs, but what counts as an acceptable resolution, and this same tension repeatedly surfaces across the cases examined so far. When observed carefully, no speaker consults a rule book before saying *paṇḍlu*, nor does anyone formally deliberate between *ceṭlu* and *ceṭṭul* in ordinary conversation. No regional assembly votes to expand *kōsam* into *kōsaram*, and no Japanese speaker consciously calculates how best to restructure *strike*. If unrelated languages independently converge on such patterned reorganisations without explicit coordination, then what differentiates among the possible alternatives, and at what stage does this differentiation occur?

7. Differentiation in the Moment of Speech

So far, we have looked at how forms survive, spread, or disappear over time. But the same logic becomes even clearer when we look at something much smaller and more immediate: the moment of speaking itself. Long before a form becomes dominant across regions or stabilises in literature, it is already being tested in real time. Every utterance places demands on coordination. Some combinations move smoothly, while others introduce friction and this difference registers instantly in the flow of speech.

Consider the contrast between Telugu constructions *evari kōsam* and *evari kōsaram* ('for whom'). Both forms exist. Both are meaningful and are defended by speakers in different regions. Yet in rapid everyday speech, *evari kōsam* settles more readily into the phrase. The shorter form fits compactly within the surrounding sequence. The longer form adds syllabic weight without altering the meaning. This difference becomes more apparent when the phrase is repeated or extended across melody, as in songs. Substituting *kōsaram* in such contexts increases syllabic weight and alters the rhythmic contour of the phrase, making it less stable under rapid articulation. The misfit is not ideological, yet it is observable in the flow of speech itself. What matters here is that speakers do not arrive at this preference by discussion or agreement. Even speakers who consciously identify with the longer form often default to the shorter form under conditions of speed or rhythm. The decision occurs before awareness has time to intervene. This kind of defaulting aligns with observations from speech production research showing that forms which integrate smoothly into ongoing articulation are more likely

to recur, while sequences that interrupt planning or execution tend not to be repeated (Levelt 1989). Importantly, this recurrence does not require conscious choice. It emerges from how utterances unfold moment by moment.

A similar phenomenon appears in Telugu expressions such as *undi kadā* and *vundi gada*. Both convey the same meaning ('it is there, isn't it?'). Both are widely understood without difficulty. Yet many speakers find *vundi gada* easier to produce in rapid speech, where the consonantal onset *vu* /v/ allows a smoother transition into the phrase than the vowel-initial *undi kadā*. Likewise, the transition from *di* to *ga* permits smoother segmental progression than the transition to *ka*. This does not imply *ga* is inherently more preferable than *ka*. In other environments, such as *ennō kānukalu* ('many gifts'), replacing the *ka* with *ga* immediately disrupts the sequence. The difference arises from local interactions within particular articulatory environments. Studies of coarticulation confirm that ease emerges from such local configurations rather than from abstract segmental preferences (Fowler 1986). This same pattern appears in contrasts in Telugu such as *ēmi cēstunnāvu* and *ēm jēstunnāvu* ('what are you doing?'). Even when the latter is dismissed as regional or informal, the shift from *cē* to *jē* allows a smoother transition within rapid speech. The resulting sequence aligns more compactly with the surrounding segments, reducing articulatory adjustment required during the phrase. While such forms may later attract social interpretations, the difference first becomes visible at the level of production, in the timing and sequencing of speech.

Across all these cases, the pattern is consistent. The outcomes do not emerge from reflection, education, or social alignment, but from pressure. Certain forms demand more from the speaker, while others demand less. Those that demand less will pass unnoticed and repeat themselves effortlessly. Those that demand more require conscious maintenance or correction. Crucially, none of these outcomes require speakers to compare forms, reflect on alternatives, or decide which variant should prevail. At this level the usual language of choice begins to lose its force. Speakers are not weighing alternatives in the way one chooses between items on a list. The differentiation takes place before such deliberation becomes possible. What feels easier is not chosen because it is preferred, but it becomes preferred because it fits the structural conditions of use.

Once this is recognized, the broader patterns discussed earlier take on a different shape. Forms that spread widely do not do so solely because speakers have promoted or defended them. Their viability is already evident in the act of speaking. Over time, what functions smoothly within repeated use accumulates wider reach, while forms that require additional structural accommodation remain confined. If this differentiation is already observable within moment-to-moment speech, then at what stage does it take place, and what governs it?

8. From Pattern to Process

Across the domains examined here, a recurrent empirical pattern has become increasingly difficult to dismiss. Forms do not simply appear and disappear at random, nor do they persist solely because speakers endorse them or institutions promote them. Across morphology,

register variation, dialect contact, lexical rivalry, borrowing, and moment-to-moment speech, multiple grammatically possible alternatives coexist, yet their distribution is not neutral. Certain configurations circulate more readily, others remain restricted, and still others undergo restructuring before stabilising.

Existing accounts have illuminated important dimensions of this process. Sociolinguistic models explain how variants acquire social meaning and circulate within communities. Usage-based and exemplar approaches demonstrate how repetition strengthens representation and how frequency conditions phonetic reduction and entrenchment. Phonetic research has shown that articulatory and perceptual pressures operate continuously in production. In the cases examined here, speakers are not typically weighing alternatives or consciously selecting outcomes at the point of use. In each domain, differentiation among co-possible forms becomes detectable prior to large-scale frequency divergence or institutional standardisation. Yet convergence nevertheless occurs. If such convergence does not depend on overt choice then deliberate selection cannot be the primary driver of the observed patterns. This points to a further question.

Taken together, these observations suggest that differentiation does not arise solely in individual preference, regional identity, or institutional alignment, nor can it be fully reduced to the cumulative strengthening of representations through frequency, as emphasised in standard usage-based and exemplar models. It must also involve processes internal to language itself. Forms do not stabilise because speakers intend them to. They stabilise because, in use, some forms consistently function more readily within the relevant structural conditions than competing alternatives. If stabilisation is structural rather than semantic or social, if grammatical generation alone does not determine circulation, if pressure-based accounts primarily explain tendencies in realisation rather than initial divergence among alternatives, and if frequency itself reflects accumulated reproduction, then the locus of differentiation must lie elsewhere in the process of language use. Accordingly, the most coherent continuation of this reasoning is to posit a mechanism operating during production that conditions which forms become repeatable in the first place. The evidence presented here suggests that alternatives are no longer functionally equivalent once enacted in use. Some integrate into ongoing speech with fewer adjustments in coordination and timing, while others impose greater structural load. These differences are observable in real-time production before they are amplified distributionally. If such differences consistently precede frequency divergence, then an account of linguistic stabilization must identify the process through which alternatives become differentially repeatable during use.

On this basis, I propose that a process operates during production and reproduction that conditions that conditions the selection of forms. I call this process internal evaluation. By internal evaluation, I do not mean conscious judgment or overt comparison among alternatives. Rather, I refer to the structured differentiation that arises as forms are enacted relative to the speaker's phonological and constructional system, where alternatives differ in how readily they integrate within its organisation. Within this system, grammatical possibilities are not neutral once realised, but differ in the degree to which they function smoothly during use.

Internal evaluation thus designates the mechanism through which co-possible forms are differentially sustained or restricted in real time. Over repeated enactment, these micro-level differences accumulate distributionally, giving rise to patterns that later appear as preference, norm, or frequency effect. In this view, frequency is not rejected as explanatory, but repositioned as a factor that consolidates patterns whose differentiation is already operative during use. This proposal does not displace grammar, phonetics, or usage-based insight. Grammar governs what can be formed. Phonetic pressures shape realisation. Usage strengthens representation. But internal evaluation identifies a prior locus at which alternatives are differentially sustained within production itself. The following discussion situates this proposal in relation to existing phonological, phonetic, and exemplar-based accounts.

9. Discussion

The perspective developed here advances a unified account of the conditions under which grammatically equivalent forms stabilise prior to frequency differentiation or representational entrenchment, a problem that has remained only partially addressed within phonological models. Rather than treating stabilisation as the outcome of conscious speaker choice or as a by-product of morphological structure, I propose internal evaluation as a decisive, enactment-based mechanism that filters grammatical possibilities in real time, unifies structural and morphological variation across registers, dialects, borrowing, and everyday speech, and relocates the primary locus of selection prior to frequency and sociological interpretation. Within this view, the differential survival of forms across contexts, the recurrence of repair as a response to misalignment, and the apparent directionality of change are understood as outcomes of a common underlying process. This shift in perspective does not eliminate social meaning or speaker agency, but instead repositions them as interpretive layers that follow structural differentiation, rather than as its primary source.

The term *evaluation* has appeared in linguistics in multiple senses. In generative traditions (Chomsky 1965), grammars were compared through evaluation metrics; in Optimality Theory (Prince & Smolensky 2004), candidate outputs are evaluated against ranked constraints; and in sociolinguistics, evaluation refers to speakers' overt commentary on variants (Labov 1972). But the use of internal evaluation proposed in this essay differs from these traditions. It does not refer to formal comparison among grammars, to constraint-ranking over abstract candidates, or to conscious metalinguistic judgment. Rather, it designates a process of differentiation operating during enactment itself, prior to reflection and independent of explicit commentary. Therefore, the term is used here not as a label for an established framework, but as a descriptive designation for the mechanism required to account for differential stabilisation among co-possible forms.

Taken together, internal evaluation can be defined as an emergent property of neuro-motor-perceptual coordination during moment-to-moment speech enactment. It does not consist in conscious comparison among alternatives, nor in post-hoc correction, nor in the application of a static hierarchy of abstract constraints. Rather, it refers to the continuous

differentiation that arises as structural configurations are produced and perceived within the speaker's currently operative coordination regime. The operative coordination regime itself is historically entrenched yet plastic, and can recalibrate under sustained ecological exposure. Although the neuro-motor-perceptual capacities underlying this process are shared across humans, the structural organisation relative to which differentiation occurs is language-specific and historically entrenched. Thus, internal evaluation is universal in mechanism but system-relative in outcome.

9.1 How it differs from Phonological and Sociolinguistic theories

Traditional phonology has long acknowledged factors such as ease of articulation and constraint satisfaction (Prince & Smolensky 2004; Hayes 2009). However, these are typically treated as rule systems or ranked constraint architectures that define well-formed outputs independently of the context of use. Browman & Goldstein (1992) likewise demonstrated that speech production is organised through coordinated gestural dynamics rather than purely symbolic rules. However, while such approaches illuminate how speech gestures are organised and timed, they do not directly explain why certain grammatical alternatives stabilise while others remain marginal in use. What remains less explored is why the same form can succeed in poetry but fail in everyday speech, or why structurally equivalent outputs exhibit divergent stabilisation across registers and domains. Viewed through the perspective developed here, these contrasts reflect differences in the conditions of enactment rather than violations of rules. The grammatical system may generate multiple possibilities, yet these possibilities are not neutral once enacted in use. In other words, this perspective argues that grammatical equality does not imply equivalence in performance.

Similarly, sociolinguistic models have offered powerful explanations for how linguistic forms acquire social meaning, how accents index identity, and how variants become associated with prestige or stigma (Silverstein 2003; Agha 2007). These models explain why speakers argue over forms and defend them as more beautiful, authentic, or correct. However, they struggle with cases in which convergence emerges without explicit ideological alignment, or where socially low-status forms nevertheless circulate robustly in everyday production. From the perspective of internal evaluation, such cases suggest that structural compatibility within the production system may condition stabilisation prior to, and independently of, overt social interpretation. Within this view, ideology does not initiate differentiation but attaches to forms whose distribution has already been shaped through repeated enactment.

9.2 How it differs from Phonetic theories

Phonetic theories have emphasised that articulatory and perceptual pressure exist and operate continuously in speech production (Lindblom 1990b; Ohala 1993). These accounts successfully explain gradient reduction, assimilation, and variation in realisation. However, they primarily model how individual segments or features are adjusted during production rather than how entire competing forms are differentially stabilised, reshaped, or restricted when

multiple grammatically possible alternatives coexist. In such accounts, the causal chain typically runs from articulatory pressure to phonetic adjustment to gradual structural change.

Internal evaluation does not deny articulatory pressure. It proposes instead that such pressures are mediated through a system-relative coordination regime during enactment. Reduction, expansion, and restructuring are not treated merely as biomechanical outcomes, but as consequences of how configurations integrate within the speaker's broader phonological and constructional organisation. Thus, the question is not simply which sequence is easier in isolation, but which form aligns with the active structural organisation of the system. Phonetic pressure may be universal, but its stabilising effect is conditional. Here such pressure becomes operative through coordination-based differentiation during enactment, which biases reproduction before frequency divergence.

9.3 Internal Evaluation and Early Bias in Usage-Based/Exemplar Accounts

Usage-based and exemplar approaches have convincingly demonstrated that phonetic and structural patterns become organised through repeated exposure and frequency-weighted representations in memory (Pierrehumbert 2001; Bybee 2010). These models do not deny that differentiation can begin prior to large-scale frequency divergence. Rather, they typically assume that small, distributed asymmetries arise from multiple sources, including pragmatic fit, semantic nuance, phonetic or processing ease, analogy, and social alignment. However, none of these factors is decisive alone. Instead, slight advantages across contexts accumulate, and frequency amplifies these differences through repetition and entrenchment. In this architecture, selection emerges from distributed weak biases whose effects become visible only once usage differentials grow large enough to sustain dominance.

Internal evaluation departs from this view not by rejecting early asymmetries, but by identifying their locus and structure more precisely. The claim is not that 'smoother articulation wins', nor that shorter forms automatically gain a marginal advantage. Rather, the claim is that alternatives are not neutral at the point of enactment. They differ in how they integrate with the speaker's active phonological and constructional organisation, including entrenched bilingual templates where relevant. If differentiation were reducible to surface compactness alone, then a structurally compact native Telugu construction such as *ōhinchalēdu kadu* would be expected to circulate more widely than its semantically equivalent hybrid construction *expect cheyaledu kada* ('you didn't expect it, did you?'). The fact that usage often favours the latter, despite its greater length, indicates that compactness alone cannot account for stabilisation. Likewise, variation in pronunciations such as 'Eye-Ran' or 'Eye-Raq', as opposed to more widely accepted realisations of *Iran* or *Iraq*, cannot be reduced to differences in syllable count or articulatory complexity. Rather, such realisations reflect alignment with the speaker's current phonological organisation. The variation does not arise from inability, but from system-relative integration. In contexts where speakers aim to approximate a different standardised or prestige pronunciation, additional monitoring or adjustment may be required, indicating that the preferred realisation reflects the default coordination pattern of the system.

In this view, differentiation is system-relative rather than merely distributed. A form may function smoothly within one phonological or constructional organisation yet introduce strain within another. This helps account for domain-specific differentiation such as the *pannu* split across lexical senses and registers, where identical surface forms diverge in stabilisation patterns. It also explains why restructuring may occur instead of simple reduction, and why equally short forms can show different trajectories of survival. The issue is not general ease, but compatibility with the organisation of the system as currently structured.

Exemplar theory accounts for long-term convergence through statistical accumulation in memory clouds, where frequency strengthens stored representations and shapes probabilistic outcomes over time. In this architecture, the causal sequence runs from stored exemplars to probabilistic bias to production choice to reinforcement. Internal evaluation reorders this sequence. It proposes that coordination-based differentiation during enactment conditions which configurations are repeatedly produced in the first place. Frequency then consolidates these patterns within representational space. Thus, differentiation is not generated by frequency, rather, frequency amplifies differentiation already operative in production. The distinction, therefore, is not between bias and no bias, but between distributed multi-factor bias and system-relative locus of differentiation. Usage-based approaches, including exemplar models, locate differentiation primarily in the cumulative effects of repetition and entrenchment. Internal evaluation locates a prior stage of system-relative filtering that operates during production itself, through which co-possible forms are differentially sustained or restricted before measurable distributional divergence emerges.

9.3.1 Comparison of the Models

The contrast between these two explanatory accounts can be illustrated schematically.

**Standard Usage-based /
Exemplar Models**

**Internal Evaluation
Model**

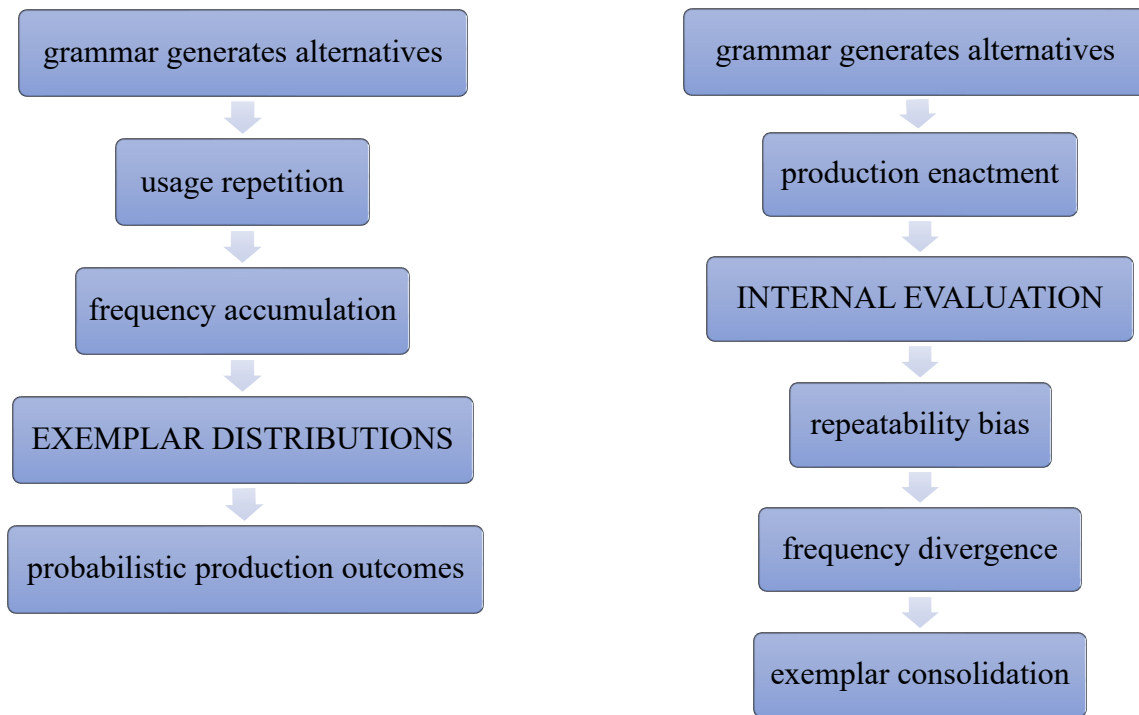


Figure 1. Schematic representation of the causal ordering in usage-based/exemplar models and the internal evaluation model. In usage-based/exemplar architecture, differentiation emerges through frequency-weighted exemplar distribution. In the internal evaluation architecture, by contrast, differentiation arises during production itself, before frequency divergence. In this sense, internal evaluation does not replace exemplar dynamics but logically precedes them, specifying the production-level conditions under which certain configurations become repeatable enough to enter exemplar distributions.

9.4 Implications

Viewed through perspective on internal evaluation, debates about which regional form is better, purer, or more correct can be reframed. Differences between forms need not be interpreted as failures of intelligence, education, or cultural legitimacy, nor solely as victories secured through political influence or social power. Rather, they may be understood as outcomes of the same internal evaluative dynamics operating under different structural conditions. This perspective suggests that forms associated with less dominant regions are not necessarily inferior variants that ‘lost’ a competition. They are outcomes that emerged, functioned, and persisted when they aligned with the coordination patterns of their speech communities. When a different form eventually became more widespread, as in the cases examined here, it may have done so because it resolved structural pressures more effectively across a broader range of contexts. This reframing does not deny the role of social or political forces, but positions them as interacting with rather than replacing, system-relative structural differentiation.

At a broader level, what unifies all the observations in this essay is the proposal of an internal evaluative process that operates during enactment. This process does not privilege region, prestige, or intention. Rather, it differentiates among alternatives relative to the

structural organisation of the system. This perspective removes neither structural constraints nor meaning. Grammar still determines what can be formed, but internal evaluation conditions what configurations are repeatedly sustained in use. Over time, such configurations may spread across speech communities and, under certain circumstances, become standardised. Crucially, this perspective does not reduce stabilisation to compression alone. While several illustrations involved structural tightening, what matters is not reduction *per se* but the resolution of structural tension within a given phonological and constructional organisation. In some contexts, this may involve compression, in others, expansion or reorganisation may provide a more stable configuration.

Beyond reframing debates about legitimacy, this perspective also clarifies why some forms remain regionally confined while others travel more widely. Languages continually generate new forms, often within specific regions or communities. These forms take the shapes they do because they fit the internal evaluative conditions of the speakers who use them, which also why those speakers often experience their local forms as especially natural or aesthetically satisfying. When such forms remain confined to their originating environments, they continue to function coherently within those structural conditions. However, when they circulate more broadly, they encounter different coordination regimes. As a result, some remain region-specific, whereas others extend beyond their point of origin. Thus regional attachment and wider convergence are not opposing forces but outcomes of the same internal evaluative process operating across different structural environments.

10. Conclusion and Future Directions:

If internal evaluation plays the role proposed here, then several consequences follow. First, forms that integrate more smoothly into ongoing coordination patterns should tend to circulate more frequently in spontaneous speech, even when multiple grammatical alternatives exist. Second, artificially introduced expanded forms should show lower rates of spontaneous adoption unless they align with existing phonological organisation. Third, in contact situations, borrowed forms should converge predictably toward the syllable structure and timing patterns of the recipient language rather than remain in structurally unsettled states. Finally, high-frequency embodied domains may exhibit stronger compression tendencies than low-frequency abstract domains, reflecting cumulative reinforcement through repeated enactment. These expectations are testable through corpus analysis, controlled production experiments, and fine-grained measurements of temporal organisation in speech. In this way, the proposal invites targeted empirical investigation of production-level differentiation prior to large-scale frequency divergence.

The goal of this essay has not been to propose a new theory of language, but to clarify the shape of a problem that grammar alone does not resolve. By following concrete cases across morphology, registers, dialect contact, lexical competition, borrowing, and real-time speech, we arrive at a position where explanations based solely on speaker choice, regional dominance, or stylistic preference appear insufficient to account for differential stabilisation among possible forms. The present account therefore advances a hypothesis about the locus at which

such differentiation may occur. Any account of language change seeking to explain differential stabilisation among co-possible forms must consider that forms are evaluated during use, as they are produced and reproduced within a speaker's coordination system, long before they are named, debated, or standardised. This internal evaluation need not be mysterious. It may operate through structured regularities, constraint-like tendencies, or other forms of implicit organisation that shape production and perception. The present paper offers an initial formulation of such a process, and a fuller account of how this evaluative process is organised, what measurable coordination signatures it produces, and how it varies across languages remains a task for future research. For now, it is sufficient to recognize that change may begin not with explicit choice, but with differential reproducibility during enactment. Sometimes, before a system can be fully explained, it must first be allowed to reveal the pressures that organise it.

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